

An Automatic Finite-Sample Robustness Metric: Can Dropping a Little Data Make a Big Difference?

Ryan Giordano (rgiordano@berkeley.edu)¹

JSM 2026: Data Selection Beyond Estimation Efficiency

¹With coauthors Rafe Meager (UNSW) and Tamara Broderick (MIT)

Dropping data: Mexico Microcredit

Example: Angelucci et al. [2015], a randomized controlled trial study of the efficacy of microcredit in Mexico based on $N=16,560$ data points. A regression was run to estimate the average effect of microcredit.

Original result: Treatment effect statistically insignificant at 95%.

Policy implication: Disinvest in microcredit initiatives.

Dropping data: Mexico Microcredit

Example: Angelucci et al. [2015], a randomized controlled trial study of the efficacy of microcredit in Mexico based on $N = 16,560$ data points. A regression was run to estimate the average effect of microcredit.

Original result: Treatment effect statistically insignificant at 95%.

Policy implication: Disinvest in microcredit initiatives.

Data dropping: Can produce both positive and negative statistically significant results dropping no more than 15 data points ($< 0.1\%$).

Policy implication: Run a higher-powered study (not just larger N).

Dropping data: Mexico Microcredit

Example: Angelucci et al. [2015], a randomized controlled trial study of the efficacy of microcredit in Mexico based on $N = 16,560$ data points. A regression was run to estimate the average effect of microcredit.

Original result: Treatment effect statistically insignificant at 95%.

Policy implication: Disinvest in microcredit initiatives.

Data dropping: Can produce both positive and negative statistically significant results dropping no more than 15 data points ($< 0.1\%$).

Policy implication: Run a higher-powered study (not just larger N).

Cannot find influential subsets by brute force!

We provide a fast, automatic tool to approximately identify the most influential set of points.

- Why and when might you care about sensitivity to data dropping?
- How do we identify influential sets? When is our method accurate?
(A formalization of the problem and the class of estimators we study.)
- Examine real-life examples of analyses: some sensitive, some not.
(The results may defy your intuition.)
- What kinds of analyses are sensitive to data dropping?
(Comparison to standard errors, gross errors, and how to mitigate.)

Dropping data: Motivation

When do you care that you can **reverse your conclusion** by removing a **small proportion** of your data?

Dropping data: Motivation

When do you care that you can **reverse your conclusion** by removing a **small proportion** of your data?

Not always! But sometimes you surely do, especially when you want to **generalize to unseen, systematically different populations**.

Suppose you have a farm, and want to know whether your average yield is > 170 bushels per acre. At harvest, you measure 200 bushels per acre.

Dropping data: Motivation

When do you care that you can **reverse your conclusion** by removing a **small proportion** of your data?

Not always! But sometimes you surely do, especially when you want to **generalize to unseen, systematically different populations**.

Suppose you have a farm, and want to know whether your average yield is > 170 bushels per acre. At harvest, you measure 200 bushels per acre.

- Scenario one: > 170 bushels per acre means you make a profit.
 - Don't care about sensitivity to small subsets.

Dropping data: Motivation

When do you care that you can **reverse your conclusion** by removing a **small proportion** of your data?

Not always! But sometimes you surely do, especially when you want to **generalize to unseen, systematically different populations**.

Suppose you have a farm, and want to know whether your average yield is > 170 bushels per acre. At harvest, you measure 200 bushels per acre.

- Scenario one: > 170 bushels per acre means you make a profit.
 - Don't care about sensitivity to small subsets.
- Scenario two: Want to recommend methods to a distant friend.
 - Might care about sensitivity to small subsets!

Dropping data: Motivation

When do you care that you can **reverse your conclusion** by removing a **small proportion** of your data?

Not always! But sometimes you surely do, especially when you want to **generalize to unseen, systematically different populations**.

Suppose you have a farm, and want to know whether your average yield is > 170 bushels per acre. At harvest, you measure 200 bushels per acre.

- Scenario one: > 170 bushels per acre means you make a profit.
 - Don't care about sensitivity to small subsets.
- Scenario two: Want to recommend methods to a distant friend.
 - Might care about sensitivity to small subsets!

Specifically, often in statistical applications:

- Policy population is different from analyzed population,
- Small fractions of data are missing not-at-random,
- We report a convenient summary (e.g. mean) of a complex effect.

Formalizing the question.

Example: Least squares

Formalizing the question.

Example: Least squares

A data point d_n has regressors x_n and response y_n : $d_n = (x_n, y_n)$.

Formalizing the question.

Example: Least squares

A data point d_n has regressors x_n and response y_n : $d_n = (x_n, y_n)$.

The estimator $\hat{\theta} \in \mathbb{R}^P$ satisfies:

$$\begin{aligned}\hat{\theta} &:= \arg \min_{\theta} \frac{1}{2} \sum_{n=1}^N (y_n - \theta^T x_n)^2 \\ \Leftrightarrow \sum_{n=1}^N (y_n - \hat{\theta}^T x_n) x_n &= 0.\end{aligned}$$

Formalizing the question.

Example: Least squares

A data point d_n has regressors x_n and response y_n : $d_n = (x_n, y_n)$.

The estimator $\hat{\theta} \in \mathbb{R}^P$ satisfies:

$$\begin{aligned}\hat{\theta} &:= \arg \min_{\theta} \frac{1}{2} \sum_{n=1}^N (y_n - \theta^T x_n)^2 \\ \Leftrightarrow \sum_{n=1}^N (y_n - \hat{\theta}^T x_n) x_n &= 0.\end{aligned}$$

Make a qualitative decision using:

- A particular component: $\hat{\theta}_k$
- The end of a confidence interval: $\hat{\theta}_k + \frac{1.96}{\sqrt{N}} \hat{\sigma}(\hat{\theta})$

Formalizing the question.

Example: Least squares

A data point d_n has regressors x_n and response y_n : $d_n = (x_n, y_n)$.

The estimator $\hat{\theta} \in \mathbb{R}^P$ satisfies:

$$\begin{aligned}\hat{\theta} &:= \arg \min_{\theta} \frac{1}{2} \sum_{n=1}^N (y_n - \theta^T x_n)^2 \\ \Leftrightarrow \sum_{n=1}^N (y_n - \hat{\theta}^T x_n) x_n &= 0.\end{aligned}$$

Make a qualitative decision using:

- A particular component: $\hat{\theta}_k$
- The end of a confidence interval: $\hat{\theta}_k + \frac{1.96}{\sqrt{N}} \hat{\sigma}(\hat{\theta})$

General setup: Z-estimators

Formalizing the question.

Example: Least squares

A data point d_n has regressors x_n and response y_n : $d_n = (x_n, y_n)$.

The estimator $\hat{\theta} \in \mathbb{R}^P$ satisfies:

$$\begin{aligned}\hat{\theta} &:= \arg \min_{\theta} \frac{1}{2} \sum_{n=1}^N (y_n - \theta^T x_n)^2 \\ \Leftrightarrow \sum_{n=1}^N (y_n - \hat{\theta}^T x_n) x_n &= 0.\end{aligned}$$

Make a qualitative decision using:

- A particular component: $\hat{\theta}_k$
- The end of a confidence interval: $\hat{\theta}_k + \frac{1.96}{\sqrt{N}} \hat{\sigma}(\hat{\theta})$

General setup: Z-estimators

We observe N data points $d_1, \dots, d_N$ (in any domain).

Formalizing the question.

Example: Least squares

A data point d_n has regressors x_n and response y_n : $d_n = (x_n, y_n)$.

The estimator $\hat{\theta} \in \mathbb{R}^P$ satisfies:

$$\begin{aligned}\hat{\theta} &:= \arg \min_{\theta} \frac{1}{2} \sum_{n=1}^N (y_n - \theta^T x_n)^2 \\ \Leftrightarrow \sum_{n=1}^N (y_n - \hat{\theta}^T x_n) x_n &= 0.\end{aligned}$$

Make a qualitative decision using:

- A particular component: $\hat{\theta}_k$
- The end of a confidence interval: $\hat{\theta}_k + \frac{1.96}{\sqrt{N}} \hat{\sigma}(\hat{\theta})$

General setup: Z-estimators

We observe N data points $d_1, \dots, d_N$ (in any domain).

The estimator $\hat{\theta} \in \mathbb{R}^P$ satisfies:

$$\sum_{n=1}^N G(\hat{\theta}, d_n) = 0_P.$$

$G(\cdot, d_n)$ is “nice,” $\mathbb{R}^P$ -valued.
E.g. MLE, MAP, VB, IV &c.

Formalizing the question.

Example: Least squares

A data point d_n has regressors x_n and response y_n : $d_n = (x_n, y_n)$.

The estimator $\hat{\theta} \in \mathbb{R}^P$ satisfies:

$$\begin{aligned}\hat{\theta} &:= \arg \min_{\theta} \frac{1}{2} \sum_{n=1}^N (y_n - \theta^T x_n)^2 \\ \Leftrightarrow \sum_{n=1}^N (y_n - \hat{\theta}^T x_n) x_n &= 0.\end{aligned}$$

Make a qualitative decision using:

- A particular component: $\hat{\theta}_k$
- The end of a confidence interval: $\hat{\theta}_k + \frac{1.96}{\sqrt{N}} \hat{\sigma}(\hat{\theta})$

General setup: Z-estimators

We observe N data points $d_1, \dots, d_N$ (in any domain).

The estimator $\hat{\theta} \in \mathbb{R}^P$ satisfies:

$$\sum_{n=1}^N G(\hat{\theta}, d_n) = 0_P.$$

$G(\cdot, d_n)$ is “nice,” $\mathbb{R}^P$ -valued.
E.g. MLE, MAP, VB, IV &c.

Make a qualitative decision using $\phi(\hat{\theta})$ for a smooth, real-valued ϕ .

(WLOG try to increase $\phi(\hat{\theta})$.)

Data dropping as data reweighting.

Question: Can we make a big change in $\phi(\hat{\theta})$ by dropping $\lfloor \alpha N \rfloor$ datapoints, for some small proportion α ?

Data dropping as data reweighting.

Question: Can we make a big change in $\phi(\hat{\theta})$ by dropping $\lfloor \alpha N \rfloor$ datapoints, for some small proportion α ? **Two big problems:**

Data dropping as data reweighting.

Question: Can we make a big change in $\phi(\hat{\theta})$ by dropping $\lfloor \alpha N \rfloor$ datapoints, for some small proportion α ? **Two big problems:**

- There are $\binom{N}{\lfloor \alpha N \rfloor}$ sets to check. (E.g. $\binom{16,560}{15} \approx 1.5 \cdot 10^{51}$)

Data dropping as data reweighting.

Question: Can we make a big change in $\phi(\hat{\theta})$ by dropping $\lfloor \alpha N \rfloor$ datapoints, for some small proportion α ? **Two big problems:**

- There are $\binom{N}{\lfloor \alpha N \rfloor}$ sets to check. (E.g. $\binom{16,560}{15} \approx 1.5 \cdot 10^{51}$)
- Evaluating $\hat{\theta}$ re-solving the estimating equation.
 - E.g., re-running OLS. ($0.001s \cdot 1.5 \cdot 10^{51} \approx 4.8 \cdot 10^{40}$ years)
 - Other examples are even harder (VB, machine learning)

Data dropping as data reweighting.

Question: Can we make a big change in $\phi(\hat{\theta})$ by dropping $\lfloor \alpha N \rfloor$ datapoints, for some small proportion α ? **Two big problems:**

- There are $\binom{N}{\lfloor \alpha N \rfloor}$ sets to check. (E.g. $\binom{16,560}{15} \approx 1.5 \cdot 10^{51}$)
- Evaluating $\hat{\theta}$ re-solving the estimating equation.
 - E.g., re-running OLS. ($0.001s \cdot 1.5 \cdot 10^{51} \approx 4.8 \cdot 10^{40}$ years)
 - Other examples are even harder (VB, machine learning)

Our idea: Smoothly approximate the effect of leaving out points.

Data dropping as data reweighting.

Question: Can we make a big change in $\phi(\hat{\theta})$ by dropping $\lfloor \alpha N \rfloor$ datapoints, for some small proportion α ? **Two big problems:**

- There are $\binom{N}{\lfloor \alpha N \rfloor}$ sets to check. (E.g. $\binom{16,560}{15} \approx 1.5 \cdot 10^{51}$)
- Evaluating $\hat{\theta}$ re-solving the estimating equation.
 - E.g., re-running OLS. ($0.001s \cdot 1.5 \cdot 10^{51} \approx 4.8 \cdot 10^{40}$ years)
 - Other examples are even harder (VB, machine learning)

Our idea: Smoothly approximate the effect of leaving out points.

We have N data points $d_1, \dots, d_N$, a quantity of interest $\phi(\cdot)$, and

$$\sum_{n=1}^N G(\hat{\theta}, d_n) = 0_P \quad .$$

Data dropping as data reweighting.

Question: Can we make a big change in $\phi(\hat{\theta})$ by dropping $\lfloor \alpha N \rfloor$ datapoints, for some small proportion α ? **Two big problems:**

- There are $\binom{N}{\lfloor \alpha N \rfloor}$ sets to check. (E.g. $\binom{16,560}{15} \approx 1.5 \cdot 10^{51}$)
- Evaluating $\hat{\theta}$ re-solving the estimating equation.
 - E.g., re-running OLS. ($0.001s \cdot 1.5 \cdot 10^{51} \approx 4.8 \cdot 10^{40}$ years)
 - Other examples are even harder (VB, machine learning)

Our idea: Smoothly approximate the effect of leaving out points.

We have N data points $d_1, \dots, d_N$, a quantity of interest $\phi(\cdot)$, and

$$\sum_{n=1}^N w_n G(\hat{\theta}(w), d_n) = 0_P \text{ for a weight vector } w \in \mathbb{R}^N.$$

Data dropping as data reweighting.

Question: Can we make a big change in $\phi(\hat{\theta})$ by dropping $\lfloor \alpha N \rfloor$ datapoints, for some small proportion α ? **Two big problems:**

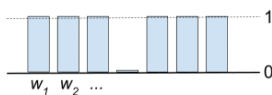
- There are $\binom{N}{\lfloor \alpha N \rfloor}$ sets to check. (E.g. $\binom{16,560}{15} \approx 1.5 \cdot 10^{51}$)
- Evaluating $\hat{\theta}$ re-solving the estimating equation.
 - E.g., re-running OLS. ($0.001s \cdot 1.5 \cdot 10^{51} \approx 4.8 \cdot 10^{40}$ years)
 - Other examples are even harder (VB, machine learning)

Our idea: Smoothly approximate the effect of leaving out points.

We have N data points $d_1, \dots, d_N$, a quantity of interest $\phi(\cdot)$, and

$$\sum_{n=1}^N w_n G(\hat{\theta}(w), d_n) = 0_P \text{ for a weight vector } w \in \mathbb{R}^N.$$

Original weights: $\vec{1} = (1, \dots, 1)$ Leave points out by setting their elements of w to zero.



Data dropping as data reweighting.

Question: Can we make a big change in $\phi(\hat{\theta})$ by dropping $\lfloor \alpha N \rfloor$ datapoints, for some small proportion α ? **Two big problems:**

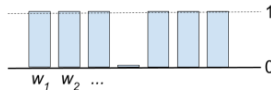
- There are $\binom{N}{\lfloor \alpha N \rfloor}$ sets to check. (E.g. $\binom{16,560}{15} \approx 1.5 \cdot 10^{51}$)
- Evaluating $\hat{\theta}$ re-solving the estimating equation.
 - E.g., re-running OLS. ($0.001s \cdot 1.5 \cdot 10^{51} \approx 4.8 \cdot 10^{40}$ years)
 - Other examples are even harder (VB, machine learning)

Our idea: Smoothly approximate the effect of leaving out points.

We have N data points $d_1, \dots, d_N$, a quantity of interest $\phi(\cdot)$, and

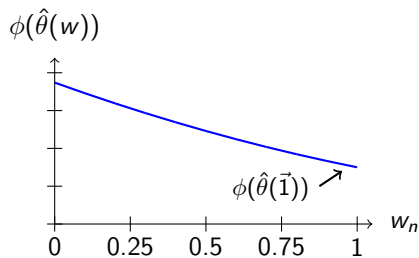
$$\sum_{n=1}^N w_n G(\hat{\theta}(w), d_n) = 0_P \text{ for a weight vector } w \in \mathbb{R}^N.$$

Original weights: $\vec{1} = (1, \dots, 1)$ Leave points out by setting their elements of w to zero.

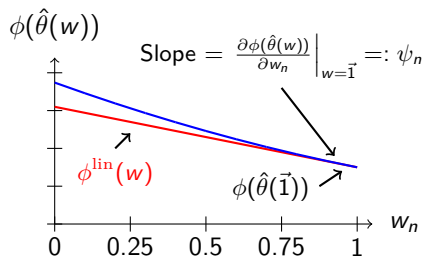


The map $w \mapsto \phi(\hat{\theta}(w))$ is well-defined even for continuous weights.

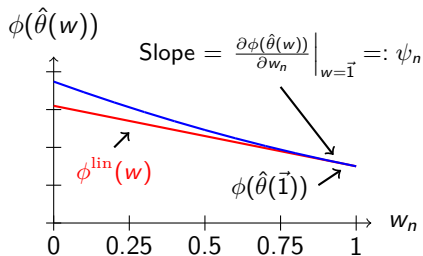
Taylor series approximation.



Taylor series approximation.

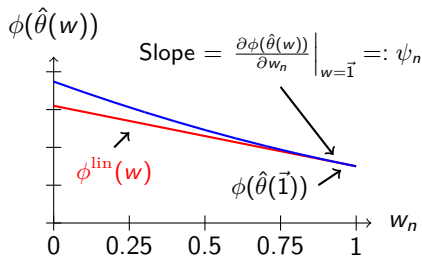


Taylor series approximation.



The values $N\psi_n$ are the **empirical influence function** [Hampel, 1986]. We call ψ_n an “influence score.”

Taylor series approximation.



The values $N\psi_n$ are the **empirical influence function** [Hampel, 1986]. We call ψ_n an “influence score.”

We can use ψ_n to form a Taylor series approximation:

$$\phi(\hat{\theta}(w)) \approx \phi^{\text{lin}}(w) := \phi(\hat{\theta}(\vec{1})) + \sum_{n=1}^N \psi_n (w_n - 1)$$

Taylor series approximation.

Problem: How much can you change $\phi(\hat{\theta}(w))$ dropping $\lfloor \alpha N \rfloor$ points?
Combinatorially hard by brute force!

Taylor series approximation.

Problem: How much can you change $\phi(\hat{\theta}(w))$ dropping $\lfloor \alpha N \rfloor$ points?
Combinatorially hard by brute force!

Approximate Problem: How much can you change $\phi^{\text{lin}}(w)$ dropping $\lfloor \alpha N \rfloor$ points? **Easy!**

$$\phi^{\text{lin}}(w) := \phi(\hat{\theta}(\vec{1})) + \sum_{n=1}^N \psi_n(w_n - 1)$$

Dropped points have $w_n - 1 = -1$. Kept points have $w_n - 1 = 0$
 $\Rightarrow$ The most influential points for $\phi^{\text{lin}}(w)$ have the most negative ψ_n .

Taylor series approximation.

Problem: How much can you change $\phi(\hat{\theta}(w))$ dropping $\lfloor \alpha N \rfloor$ points?
Combinatorially hard by brute force!

Approximate Problem: How much can you change $\phi^{\text{lin}}(w)$ dropping $\lfloor \alpha N \rfloor$ points? **Easy!**

$$\phi^{\text{lin}}(w) := \phi(\hat{\theta}(\vec{1})) + \sum_{n=1}^N \psi_n(w_n - 1)$$

Dropped points have $w_n - 1 = -1$. Kept points have $w_n - 1 = 0$
 $\Rightarrow$ The most influential points for $\phi^{\text{lin}}(w)$ have the most negative ψ_n .

Our procedure: (see [rgiordan/zaminfluence](#) on github)

- 1 Compute your original estimator $\hat{\theta}(\vec{1})$.
- 2 Compute and sort the influence scores $\psi_{(1)}, \dots, \psi_{(N)}$.
- 3 Check if $-\sum_{n=1}^{\lfloor \alpha N \rfloor} \psi_{(n)}$ is large enough to change your conclusions.

Taylor series approximation.

Problem: How much can you change $\phi(\hat{\theta}(w))$ dropping $\lfloor \alpha N \rfloor$ points?
Combinatorially hard by brute force!

Approximate Problem: How much can you change $\phi^{\text{lin}}(w)$ dropping $\lfloor \alpha N \rfloor$ points? **Easy!**

$$\phi^{\text{lin}}(w) := \phi(\hat{\theta}(\vec{1})) + \sum_{n=1}^N \psi_n(w_n - 1)$$

Dropped points have $w_n - 1 = -1$. Kept points have $w_n - 1 = 0$
 $\Rightarrow$ The most influential points for $\phi^{\text{lin}}(w)$ have the most negative ψ_n .

Our procedure: (see [rgiordan/zaminfluence](#) on github)

- 1 Compute your original estimator $\hat{\theta}(\vec{1})$.
- 2 Compute and sort the influence scores $\psi_{(1)}, \dots, \psi_{(N)}$.
- 3 Check if $-\sum_{n=1}^{\lfloor \alpha N \rfloor} \psi_{(n)}$ is large enough to change your conclusions.

How to compute the ψ_n 's? And how accurate is the approximation?

How to compute the influence scores?

How can we compute the influence scores $\psi_n = \left. \frac{\partial \phi(\hat{\theta}(w))}{\partial w_n} \right|_{w=\vec{1}}$?

How to compute the influence scores?

How can we compute the influence scores $\psi_n = \left. \frac{\partial \phi(\hat{\theta}(w))}{\partial w_n} \right|_{w=\vec{1}}$?

By the **chain rule**, $\psi_n = \left. \frac{\partial \phi(\theta)}{\partial \theta} \right|_{\hat{\theta}(\vec{1})} \left. \frac{\partial \hat{\theta}(w)}{\partial w_n} \right|_{w=\vec{1}}$.

How to compute the influence scores?

How can we compute the influence scores $\psi_n = \left. \frac{\partial \phi(\hat{\theta}(w))}{\partial w_n} \right|_{w=\vec{1}}$?

By the **chain rule**, $\psi_n = \left. \frac{\partial \phi(\theta)}{\partial \theta} \right|_{\hat{\theta}(\vec{1})} \left. \frac{\partial \hat{\theta}(w)}{\partial w_n} \right|_{w=\vec{1}}$.

Recall that $\sum_{n=1}^N w_n G(\hat{\theta}(w), d_n) = 0_P$ for all w near $\vec{1}$.

$\Rightarrow$ By the **implicit function theorem**, we can write $\left. \frac{\partial \hat{\theta}(w)}{\partial w_n} \right|_{w=\vec{1}}$ as a linear system involving $G(\cdot, \cdot)$ and its derivatives.

How to compute the influence scores?

How can we compute the influence scores $\psi_n = \left. \frac{\partial \phi(\hat{\theta}(w))}{\partial w_n} \right|_{w=\vec{1}}$?

By the **chain rule**, $\psi_n = \left. \frac{\partial \phi(\theta)}{\partial \theta} \right|_{\hat{\theta}(\vec{1})} \left. \frac{\partial \hat{\theta}(w)}{\partial w_n} \right|_{w=\vec{1}}$.

Recall that $\sum_{n=1}^N w_n G(\hat{\theta}(w), d_n) = 0_P$ for all w near $\vec{1}$.

$\Rightarrow$ By the **implicit function theorem**, we can write $\left. \frac{\partial \hat{\theta}(w)}{\partial w_n} \right|_{w=\vec{1}}$ as a linear system involving $G(\cdot, \cdot)$ and its derivatives.

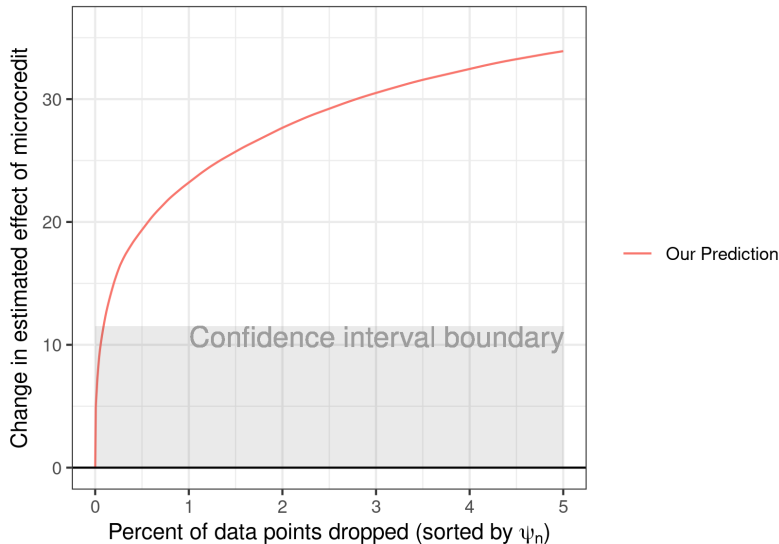
$\Rightarrow$ The ψ_n are automatically computable from $\hat{\theta}(\vec{1})$ and software implementations of $G(\cdot, \cdot)$ and $\phi(\cdot)$ using **automatic differentiation**.

```
> import jax
> import jax.numpy as np
> def phi(theta):
>     ... computations using np and theta ...
>     return value
>
> # Exact gradient of phi (first term in the chain rule above):
> jax.grad(phi)(theta_opt)
```

See [rgiordan/vittles](https://github.com/rgiordan/vittles) on github.

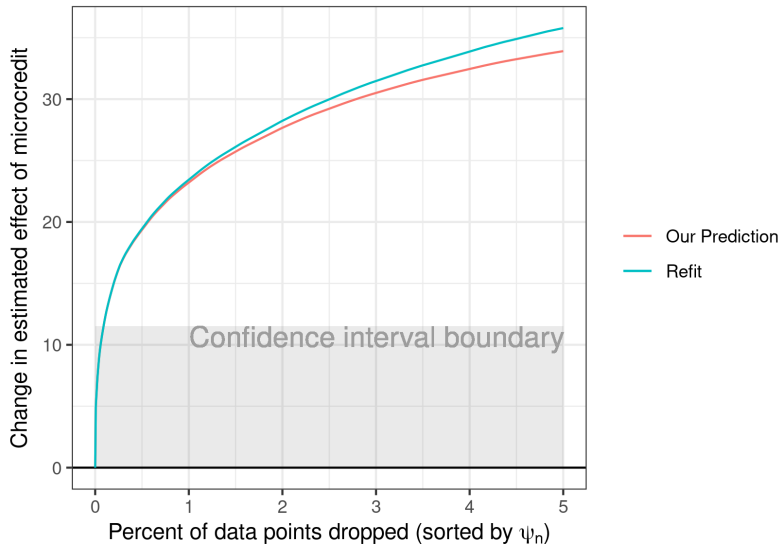
How accurate is the approximation?

Checking the approximation for Mexico microcredit.



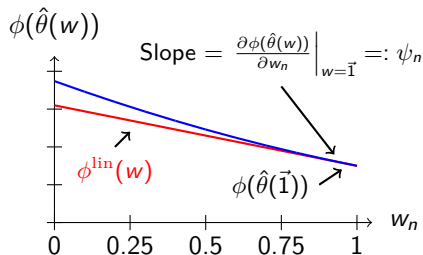
How accurate is the approximation?

Checking the approximation for Mexico microcredit.



How accurate is the approximation?

By controlling the curvature, we can control the error in the linear approximation.

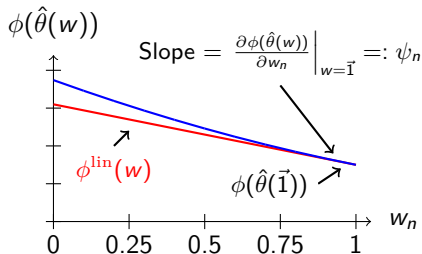


We provide **finite-sample theory** [Giordano et al., 2019b] showing that

$$\left| \phi(\hat{\theta}(w)) - \phi^{\text{lin}}(w) \right| = O \left(\left\| \frac{1}{N}(w - \vec{1}) \right\|_2^2 \right) = O(\alpha) \text{ as } \alpha \rightarrow 0.$$

How accurate is the approximation?

By controlling the curvature, we can control the error in the linear approximation.



We provide **finite-sample theory** [Giordano et al., 2019b] showing that $\left| \phi(\hat{\theta}(w)) - \phi^{\text{lin}}(w) \right| = O \left(\left\| \frac{1}{N}(w - \vec{1}) \right\|_2^2 \right) = O(\alpha)$ as $\alpha \rightarrow 0$.

But you don't need to rely on the theory to show non-robustness!

Our method returns which points to drop. **Re-running once** without those points provides an **exact lower bound** on the worst-case sensitivity.

Selected experimental results.

Original estimate (SE)	Refit estimate (SE)	Observations dropped
-4.549 (5.879)	7.030 (2.549)*	15 = 0.09%

Table: Microcredit Mexico results ($N = 16560$) [Angelucci et al., 2015].

A * indicates statistical significance at the 95% level.

Selected experimental results.

Original estimate (SE)	Refit estimate (SE)	Observations dropped
-4.549 (5.879)	7.030 (2.549)*	15 = 0.09%

Table: Microcredit Mexico results ($N = 16560$) [Angelucci et al., 2015].

Original estimate (SE)	Refit estimate (SE)	Observations dropped
33.861 (4.468)*	-9.416 (3.296)*	986 = 9.37%

Table: Cash transfers results ($N = 10518$) [Angelucci and De Giorgi, 2009]

A * indicates statistical significance at the 95% level.

Selected experimental results.

Original estimate (SE)	Refit estimate (SE)	Observations dropped
-4.549 (5.879)	7.030 (2.549)*	15 = 0.09%

Table: Microcredit Mexico results (N = 16560) [Angelucci et al., 2015].

Original estimate (SE)	Refit estimate (SE)	Observations dropped
33.861 (4.468)*	-9.416 (3.296)*	986 = 9.37%

Table: Cash transfers results (N = 10518) [Angelucci and De Giorgi, 2009]

Original estimate (SE)	Refit estimate (SE)	Observations dropped
0.029 (0.005)*	-0.009 (0.004)*	224 = 0.96%

Table: Medicaid profit results (N = 23361) [Finkelstein et al., 2012]

A * indicates statistical significance at the 95% level.

What makes an analysis sensitive? Preliminaries

We are **robust to data dropping** if, for the Δ that changes conclusions and w^* dropping the $\lfloor \alpha N \rfloor$ most influential points,

$$\Delta \geq \phi^{\text{lin}}(w^*) - \phi(\hat{\theta}(\vec{1}))$$

What makes an analysis sensitive? Preliminaries

We are **robust to data dropping** if, for the Δ that changes conclusions and w^* dropping the $\lfloor \alpha N \rfloor$ most influential points,

$$\Delta \geq \phi^{\text{lin}}(w^*) - \phi(\hat{\theta}(\vec{1}))$$

- The “signal” Δ is the smallest change that reverses your conclusion

What makes an analysis sensitive? Preliminaries

We are **robust to data dropping** if, for the Δ that changes conclusions and w^* dropping the $\lfloor \alpha N \rfloor$ most influential points,

$$\Delta \geq \phi^{\text{lin}}(w^*) - \phi(\hat{\theta}(\vec{1})) =: \hat{\sigma}_{\phi} \hat{\mathcal{S}}_{\alpha}$$

- The “signal” Δ is the smallest change that reverses your conclusion

What makes an analysis sensitive? Preliminaries

We are **robust to data dropping** if, for the Δ that changes conclusions and w^* dropping the $\lfloor \alpha N \rfloor$ most influential points,

$$\Delta \geq \phi^{\text{lin}}(w^*) - \phi(\hat{\theta}(\vec{1})) =: \hat{\sigma}_\phi \hat{\mathcal{I}}_\alpha$$

- The “signal” Δ is the smallest change that reverses your conclusion
- The “noise” $\hat{\sigma}_\phi^2 \rightarrow \lim_{N \rightarrow \infty} \text{Var}(\sqrt{N}\phi)$ (“sandwich” variance estimator)

What makes an analysis sensitive? Preliminaries

We are **robust to data dropping** if, for the Δ that changes conclusions and w^* dropping the $\lfloor \alpha N \rfloor$ most influential points,

$$\Delta \geq \phi^{\text{lin}}(w^*) - \phi(\hat{\theta}(\vec{1})) =: \hat{\sigma}_\phi \hat{\mathcal{J}}_\alpha$$

- The “signal” Δ is the smallest change that reverses your conclusion
- The “noise” $\hat{\sigma}_\phi^2 \rightarrow \lim_{N \rightarrow \infty} \text{Var}(\sqrt{N}\phi)$ (“sandwich” variance estimator)
- The “shape” $\hat{\mathcal{J}}_\alpha \rightarrow$ a nonzero constant and is $\leq \sqrt{\alpha(1-\alpha)}$

What makes an analysis sensitive? Preliminaries

We are **robust to data dropping** if, for the Δ that changes conclusions and w^* dropping the $\lfloor \alpha N \rfloor$ most influential points,

$$\Delta \geq \phi^{\text{lin}}(w^*) - \phi(\hat{\theta}(\vec{1})) =: \hat{\sigma}_\phi \hat{\mathcal{J}}_\alpha \quad \Leftrightarrow \quad \frac{\Delta}{\hat{\sigma}_\phi} \geq \hat{\mathcal{J}}_\alpha.$$

- The “signal” Δ is the smallest change that reverses your conclusion
- The “noise” $\hat{\sigma}_\phi^2 \rightarrow \lim_{N \rightarrow \infty} \text{Var}(\sqrt{N}\phi)$ (“sandwich” variance estimator)
- The “shape” $\hat{\mathcal{J}}_\alpha \rightarrow$ a nonzero constant and is $\leq \sqrt{\alpha(1-\alpha)}$

The **signal to noise ratio** $\frac{\Delta}{\hat{\sigma}_\phi}$ determines robustness to data dropping ...

What makes an analysis sensitive? Preliminaries

We are **robust to data dropping** if, for the Δ that changes conclusions and w^* dropping the $\lfloor \alpha N \rfloor$ most influential points,

$$\Delta \geq \phi^{\text{lin}}(w^*) - \phi(\hat{\theta}(\vec{1})) =: \hat{\sigma}_\phi \hat{\mathcal{J}}_\alpha \quad \Leftrightarrow \quad \frac{\Delta}{\hat{\sigma}_\phi} \geq \hat{\mathcal{J}}_\alpha.$$

- The “signal” Δ is the smallest change that reverses your conclusion
- The “noise” $\hat{\sigma}_\phi^2 \rightarrow \lim_{N \rightarrow \infty} \text{Var}(\sqrt{N}\phi)$ (“sandwich” variance estimator)
- The “shape” $\hat{\mathcal{J}}_\alpha \rightarrow$ a nonzero constant and is $\leq \sqrt{\alpha(1-\alpha)}$

The **signal to noise ratio** $\frac{\Delta}{\hat{\sigma}_\phi}$ determines robustness to data dropping ...

Contrast with sampling variability.

A 95% CI is given by $\phi(\hat{\theta}(\vec{1})) \pm \frac{1.96}{\sqrt{N}} \hat{\sigma}_\phi$. We reject $\phi(\hat{\theta}(\vec{1})) + \Delta$ when

$$\phi(\hat{\theta}(\vec{1})) + \Delta \geq \phi(\hat{\theta}(\vec{1})) + \frac{1.96}{\sqrt{N}} \hat{\sigma}_\phi$$

What makes an analysis sensitive? Preliminaries

We are **robust to data dropping** if, for the Δ that changes conclusions and w^* dropping the $\lfloor \alpha N \rfloor$ most influential points,

$$\Delta \geq \phi^{\text{lin}}(w^*) - \phi(\hat{\theta}(\vec{1})) =: \hat{\sigma}_\phi \hat{\mathcal{J}}_\alpha \quad \Leftrightarrow \quad \frac{\Delta}{\hat{\sigma}_\phi} \geq \hat{\mathcal{J}}_\alpha.$$

- The “signal” Δ is the smallest change that reverses your conclusion
- The “noise” $\hat{\sigma}_\phi^2 \rightarrow \lim_{N \rightarrow \infty} \text{Var}(\sqrt{N}\phi)$ (“sandwich” variance estimator)
- The “shape” $\hat{\mathcal{J}}_\alpha \rightarrow$ a nonzero constant and is $\leq \sqrt{\alpha(1-\alpha)}$

The **signal to noise ratio** $\frac{\Delta}{\hat{\sigma}_\phi}$ determines robustness to data dropping **and** sampling variability, but with **different thresholds**.

Contrast with sampling variability.

A 95% CI is given by $\phi(\hat{\theta}(\vec{1})) \pm \frac{1.96}{\sqrt{N}} \hat{\sigma}_\phi$. We reject $\phi(\hat{\theta}(\vec{1})) + \Delta$ when

$$\phi(\hat{\theta}(\vec{1})) + \Delta \geq \phi(\hat{\theta}(\vec{1})) + \frac{1.96}{\sqrt{N}} \hat{\sigma}_\phi \quad \Leftrightarrow \quad \frac{\Delta}{\hat{\sigma}_\phi} \geq \frac{1.96}{\sqrt{N}}.$$

What makes an analysis sensitive?

Robust to data dropping:
("dropping robustness")

$$\text{SNR} := \frac{\Delta}{\hat{\sigma}_\phi} \geq \hat{\mathcal{J}}_\alpha$$

Robust to sampling variation:
("sampling robustness")

$$\text{SNR} := \frac{\Delta}{\hat{\sigma}_\phi} \geq \frac{1.96}{\sqrt{N}}$$

What makes an analysis sensitive?

Robust to data dropping:
("dropping robustness")

$$\text{SNR} := \frac{\Delta}{\hat{\sigma}_\phi} \geq \hat{\mathcal{J}}_\alpha$$

Robust to sampling variation:
("sampling robustness")

$$\text{SNR} := \frac{\Delta}{\hat{\sigma}_\phi} \geq \frac{1.96}{\sqrt{N}}$$

-
- **Dropping robustness $\neq$ sampling robustness in general.**

Proof: $\hat{\mathcal{J}}_\alpha \neq \frac{1.96}{\sqrt{N}}$.

What makes an analysis sensitive?

Robust to data dropping:
("dropping robustness")

$$\text{SNR} := \frac{\Delta}{\hat{\sigma}_\phi} \geq \hat{\mathcal{J}}_\alpha$$

Robust to sampling variation:
("sampling robustness")

$$\text{SNR} := \frac{\Delta}{\hat{\sigma}_\phi} \geq \frac{1.96}{\sqrt{N}}$$

-
- **Dropping robustness $\neq$ sampling robustness in general.**

Proof: $\hat{\mathcal{J}}_\alpha \neq \frac{1.96}{\sqrt{N}}$.

- **When the SNR is small, sufficiently large N produces sampling robustness, but not necessarily dropping robustness.**

Proof: $\frac{1.96}{\sqrt{N}} \rightarrow 0$, but $\hat{\mathcal{J}}_\alpha \rightarrow$ a nonzero constant.

What makes an analysis sensitive?

Robust to data dropping:
("dropping robustness")

$$\text{SNR} := \frac{\Delta}{\hat{\sigma}_\phi} \geq \hat{\mathcal{J}}_\alpha$$

Robust to sampling variation:
("sampling robustness")

$$\text{SNR} := \frac{\Delta}{\hat{\sigma}_\phi} \geq \frac{1.96}{\sqrt{N}}$$

-
- **Dropping robustness $\neq$ sampling robustness in general.**

Proof: $\hat{\mathcal{J}}_\alpha \neq \frac{1.96}{\sqrt{N}}$.

- **When the SNR is small, sufficiently large N produces sampling robustness, but not necessarily dropping robustness.**

Proof: $\frac{1.96}{\sqrt{N}} \rightarrow 0$, but $\hat{\mathcal{J}}_\alpha \rightarrow$ a nonzero constant.

- **Statistical insignificance is dropping non-robust for large N .**

Proof: Insignificance means $|\phi(\hat{\theta}(\vec{1}))| \leq \frac{1.96}{\sqrt{N}} \hat{\sigma}_\phi$.

$\Rightarrow$ A result can be made significant by a change of no more than $\frac{1.96}{\sqrt{N}} \hat{\sigma}_\phi$.

$\Rightarrow$ The SNR for a conclusion of "insignificance" is $\frac{\Delta}{\hat{\sigma}_\phi} \leq \frac{1.96}{\sqrt{N}} \rightarrow 0 \leq \hat{\mathcal{J}}_\alpha$.

What makes an analysis sensitive?

Robust to data dropping:
("dropping robustness")

$$\text{SNR} := \frac{\Delta}{\hat{\sigma}_\phi} \geq \hat{\mathcal{J}}_\alpha$$

Robust to sampling variation:
("sampling robustness")

$$\text{SNR} := \frac{\Delta}{\hat{\sigma}_\phi} \geq \frac{1.96}{\sqrt{N}}$$

-
- **Dropping robustness $\neq$ sampling robustness in general.**

Proof: $\hat{\mathcal{J}}_\alpha \neq \frac{1.96}{\sqrt{N}}$.

- **When the SNR is small, sufficiently large N produces sampling robustness, but not necessarily dropping robustness.**

Proof: $\frac{1.96}{\sqrt{N}} \rightarrow 0$, but $\hat{\mathcal{J}}_\alpha \rightarrow$ a nonzero constant.

- **Statistical insignificance is dropping non-robust for large N .**

Proof: Insignificance means $|\phi(\hat{\theta}(\vec{1}))| \leq \frac{1.96}{\sqrt{N}} \hat{\sigma}_\phi$.

$\Rightarrow$ A result can be made significant by a change of no more than $\frac{1.96}{\sqrt{N}} \hat{\sigma}_\phi$.

$\Rightarrow$ The SNR for a conclusion of "insignificance" is $\frac{\Delta}{\hat{\sigma}_\phi} \leq \frac{1.96}{\sqrt{N}} \rightarrow 0 \leq \hat{\mathcal{J}}_\alpha$.

- **P-hacking is dropping non-robust for large N .**

Proof: P-hacked effect sizes are of the order $\frac{1.96}{\sqrt{N}} \hat{\sigma}_\phi$.

What makes an analysis sensitive?

Robust to data dropping:
("dropping robustness")

$$\text{SNR} := \frac{\Delta}{\hat{\sigma}_{\phi}} \geq \hat{\mathcal{J}}_{\alpha}$$

Robust to gross errors:
("gross error robustness")

Gross outliers cannot produce
arbitrarily large changes to ϕ .

What makes an analysis sensitive?

Robust to data dropping:
("dropping robustness")

$$\text{SNR} := \frac{\Delta}{\hat{\sigma}_{\phi}} \geq \hat{\mathcal{J}}_{\alpha}$$

Robust to gross errors:
("gross error robustness")

Gross outliers cannot produce arbitrarily large changes to ϕ .

-
- **Dropping non-robustness is not driven by misspecification.**

Proof: Small Δ are dropping non-robust irrespective of specification.

What makes an analysis sensitive?

Robust to data dropping:
("dropping robustness")

$$\text{SNR} := \frac{\Delta}{\hat{\sigma}_\phi} \geq \hat{\mathcal{J}}_\alpha$$

Robust to gross errors:
("gross error robustness")

Gross outliers cannot produce arbitrarily large changes to ϕ .

-
- **Dropping non-robustness is not driven by misspecification.**

Proof: Small Δ are dropping non-robust irrespective of specification.

- **Gross outliers primarily affect dropping robustness through $\hat{\sigma}_\phi$.**

Proof: For a fixed $\hat{\sigma}_\phi$, outliers decrease $\hat{\mathcal{J}}_\alpha$. (Details in paper.)

What makes an analysis sensitive?

Robust to data dropping:
("dropping robustness")

$$\text{SNR} := \frac{\Delta}{\hat{\sigma}_\phi} \geq \mathcal{J}_\alpha$$

Robust to gross errors:
("gross error robustness")

Gross outliers cannot produce arbitrarily large changes to ϕ .

-
- **Dropping non-robustness is not driven by misspecification.**

Proof: Small Δ are dropping non-robust irrespective of specification.

- **Gross outliers primarily affect dropping robustness through $\hat{\sigma}_\phi$.**

Proof: For a fixed $\hat{\sigma}_\phi$, outliers decrease $\hat{\mathcal{J}}_\alpha$. (Details in paper.)

Dropping robustness should **augment** other forms of robustness.

How to make an analysis less sensitive?

Robust to data dropping:
("dropping robustness")

$$\text{SNR} := \frac{\Delta}{\hat{\sigma}_\phi} \geq \hat{\mathcal{J}}_\alpha$$

To achieve dropping robustness, reduce $\hat{\sigma}_\phi$ and / or increase Δ .

Proof: Across typical distributions, $\hat{\mathcal{J}}_\alpha$ varies little. (Details in paper.)

How to make an analysis less sensitive?

Robust to data dropping:
("dropping robustness")

$$\text{SNR} := \frac{\Delta}{\hat{\sigma}_{\phi}} \geq \hat{\mathcal{J}}_{\alpha}$$

To achieve dropping robustness, reduce $\hat{\sigma}_{\phi}$ and / or increase Δ .

Proof: Across typical distributions, $\hat{\mathcal{J}}_{\alpha}$ varies little. (Details in paper.)

In the Mexico microcredit example,

$$\hat{\sigma}_{\phi} = 757.8 \qquad \phi(\hat{\theta}(\vec{1})) = -4.55 \qquad N = 16,560$$

The study overcame a very low signal to noise ratio with a very large N .

This (canonical) response to low signal to noise ratio — to gather more data — produces small SEs, but cannot produce dropping robustness.

- You may be concerned if you could reverse your conclusion by removing a small proportion of your data.

Conclusion

- You may be concerned if you could reverse your conclusion by removing a small proportion of your data.
- We can quickly and automatically find an approximate influential set which is accurate for small sets.

- You may be concerned if you could reverse your conclusion by removing a small proportion of your data.
- We can quickly and automatically find an approximate influential set which is accurate for small sets.
- Data dropping robustness is principally determined by the signal to noise ratio, and captures sensitivity distinct from sampling and gross error sensitivity.

Links and references

Tamara Broderick, Ryan Giordano, Rachael Meager (alphabetical authors)
“An Automatic Finite-Sample Robustness Metric: Can Dropping a Little Data Change Conclusions?”

<https://arxiv.org/abs/2011.14999>

Select blog posts with more details: <https://rgiordan.github.io>

- Data dropping sensitivity overcomes p-hacking
- Collinearity in OLS after dropping
- Influence functions and sums
- Connections to the bootstrap

Related software on github:

- [rgiordan/zaminfluence](#) (for R)
- [rgiordan/vittles](#) (for Python)

Some of my work on other forms of robustness:

- Prior sensitivity in Bayesian nonparametrics [Giordano et al., 2021]
- Approximate cross-validation (and other reweightings) [Giordano et al., 2019b,a]
- Covariances and prior sensitivity for mean field VB [Giordano et al., 2015, 2018]
- Model sensitivity of MCMC output [Giordano et al., 2018]
- Frequentist variances of MCMC posteriors (in progress)

References

- M. Angelucci and G. De Giorgi. Indirect effects of an aid program: How do cash transfers affect ineligible's consumption? *American Economic Review*, 99(1):486–508, 2009.
- M. Angelucci, D. Karlan, and J. Zinman. Microcredit impacts: Evidence from a randomized microcredit program placement experiment by Compartamos Banco. *American Economic Journal: Applied Economics*, 7(1):151–82, 2015.
- D. Belsley, E. Kuh, and R. Welsch. *Regression diagnostics: Identifying influential data and sources of collinearity*. John Wiley & Sons, 2005.
- P. Bickel, C. Klaassen, Y. Ritov, and J. Wellner. *Efficient and adaptive estimation for semiparametric models*, volume 4. Johns Hopkins University Press Baltimore, 1993.
- D. Cook. Detection of influential observation in linear regression. *Technometrics*, 19(1):15–18, 1977.
- A. Finkelstein, S. Taubman, B. Wright, M. Bernstein, J. Gruber, J. Newhouse, H. Allen, K. Baicker, and Oregon Health Study Group. The Oregon health insurance experiment: Evidence from the first year. *The Quarterly Journal of Economics*, 127(3):1057–1106, 2012.
- R. Giordano, T. Broderick, and M. I. Jordan. Linear response methods for accurate covariance estimates from mean field variational Bayes. *Advances in Neural Information Processing Systems*, 28:1441–1449, 2015.
- R. Giordano, T. Broderick, and M. I. Jordan. Covariances, robustness and variational Bayes. *The Journal of Machine Learning Research*, 19(1):1981–2029, 2018.
- R. Giordano, M. I. Jordan, and T. Broderick. A higher-order Swiss army infinitesimal jackknife. *arXiv preprint arXiv:1907.12116*, 2019a.
- R. Giordano, W. Stephenson, R. Liu, M. I. Jordan, and T. Broderick. A Swiss army infinitesimal jackknife. In *The 22nd International Conference on Artificial Intelligence and Statistics*, pages 1139–1147. PMLR, 2019b.
- R. Giordano, R. Liu, M. I. Jordan, and T. Broderick. Evaluating sensitivity to the stick-breaking prior in Bayesian nonparametrics. 2021.
- F. Hampel. The influence curve and its role in robust estimation. *Journal of the American Statistical Association*, 69(346):383–393, 1974.
- F. Hampel. *Robust statistics: The approach based on influence functions*, volume 196. Wiley-Interscience, 1986.
- P. Huber. *Robust Statistics*. John Wiley & Sons, New York, 1981.
- P. Huber and D. Donoho. The notion of breakdown point. *A Festschrift for Erich L. Lehmann*, 1983.
- R. Mises. On the asymptotic distribution of differentiable statistical functions. *The Annals of Mathematical Statistics*, 18(3):309–348, 1947.
- J. Reeds. *On the definition of von Mises functionals*. PhD thesis, Statistics, Harvard University, 1976.

Extra slides

Selected related classical ideas

Our work is most closely related to “infinitesimal robustness” [Hampel, 1974, 1986], which is based on the influence function, the first term in the Taylor series expansion of functionals of a distribution function.

- We diagnose an existing analysis rather than designing estimators.
- We use autodiff to automate the influence function computation.
- We relate sensitivity to the signal-to-noise ratio, improving intuition.

Our work is also related to the “global robustness” concept of a “breakdown point” [Huber, 1981, Huber and Donoho, 1983].

- We consider data dropping and a particular qualitative conclusion, rather than arbitrarily large changes.
- We use infinitesimal robustness for automaticity and generality, but recover non-local robustness properties by controlling the residual.

Our work shares some formal similarity with outlier detection [Cook, 1977, Belsley et al., 2005].

- We consider $O(N)$ -sized sets rather than individual points.
- We study a specific intervention, not vaguely defined “outliers.”
- We consider settings far beyond regression.

A simulation

For $N = 5,000$ data points, compute the OLS estimator from:

Regressors
 $x_n \sim \mathcal{N}(0, \sigma_x^2)$

Residuals
 $\varepsilon_n \sim \mathcal{N}(0, \sigma_\varepsilon^2)$

Responses
 $y_n = 0.5x_n + \varepsilon_n$

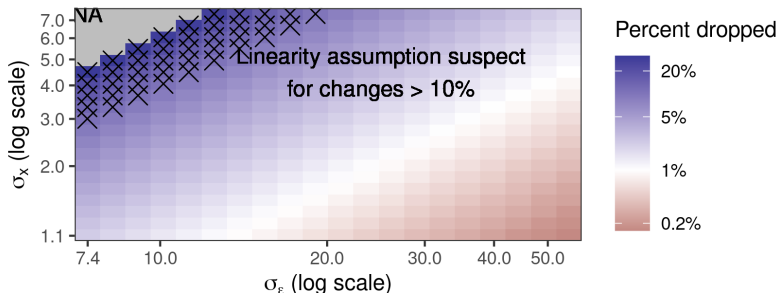


Figure: The approximate perturbation inducing proportion at differing values of σ_x and σ_ε . Red colors indicate datasets whose sign can be predicted to change when dropping less than 1% of datapoints. The grey areas indicate $\hat{\Psi}_\alpha = \text{NA}$, a failure of the linear approximation to locate any way to change the sign.

Influence function

The present work is based on the *empirical influence function*. Consider:

- True, unknown distribution function $F_\infty(x) = p(X \leq x)$
- Empirical distribution function $\hat{F}(x) = \frac{1}{N} \sum_{n=1}^N \mathbb{I}(x_n \leq x)$
- A statistical functional $T(F)$.

Influence function

The present work is based on the *empirical influence function*. Consider:

- True, unknown distribution function $F_\infty(x) = p(X \leq x)$
- Empirical distribution function $\hat{F}(x) = \frac{1}{N} \sum_{n=1}^N \mathbb{I}(x_n \leq x)$
- A statistical functional $T(F)$.

We estimate with $T(F_\infty)$ with $T(\hat{F})$.

Sample means are an example:

$$T(F) := \int x F(dx).$$

Z-estimators are, too:

$$T(F) := \theta \text{ such that } \int G(\theta, x) F(dx) = 0.$$

Influence function

The present work is based on the *empirical influence function*. Consider:

- True, unknown distribution function $F_\infty(x) = p(X \leq x)$
- Empirical distribution function $\hat{F}(x) = \frac{1}{N} \sum_{n=1}^N \mathbb{I}(x_n \leq x)$
- A statistical functional $T(F)$.

Form an (infinite-dimensional) Taylor series expansion at some F_0 :

$$T(F) = T(F_0) + T'(F_0)(F - F_0) + \text{residual}.$$

When the derivative operator takes the form of an integral

$$T'(F_0)\Delta = \int \psi(x; F_0)\Delta(dx)$$

then $\psi(x; F_0)$ is known as the *influence function*.

Where to form the expansion? There are at least two reasonable choices:

- The limiting influence function $\psi(x, F_\infty)$
- The empirical influence function $\psi(x, \hat{F})$

Influence function

- The limiting influence function (LIF) $\psi(x, F_\infty)$
 - Used in a lot of classical statistics [Mises, 1947, Huber, 1981, Hampel, 1986, Bickel et al., 1993]
 - Unobserved, asymptotic
 - Requires careful functional analysis [Reeds, 1976]
- The empirical influence function (EIF) $\psi(x, \hat{F})$
 - The basis of the present work (also [Giordano et al., 2019b,a])
 - Computable, finite-sample
 - Requires only finite-dimensional calculus

Typically the *semantics* of the EIF derive from study of the LIF.

Example: $\frac{1}{N} \sum_{n=1}^N (N\psi_n)^2 \approx \text{Var} \left(\sqrt{N}\phi(\hat{\theta}) \right).$

But the EIF measures what happens when you perturb the data at hand.

Other data perturbations will admit an analysis similar to ours!

Local robustness

The present work is an application of *local robustness*. Consider:

- Model parameter λ (e.g., data weights $\lambda = w$)
- Set of plausible models $\mathcal{S}_\lambda$ (e.g. $\mathcal{S}_\lambda = W_\alpha$)
- Estimator $\hat{\theta}(x, \lambda)$ for data x and $\lambda \in \mathcal{S}_\lambda$ (e.g. a Z-estimator)

Global robustness: $\left(\inf_{\lambda \in \mathcal{S}_\lambda} \hat{\theta}(x, \lambda), \sup_{\lambda \in \mathcal{S}_\lambda} \hat{\theta}(x, \lambda) \right)$ (Hard in general!)

Local robustness: $\left(\inf_{\lambda \in \mathcal{S}_\lambda} \hat{\theta}^{lin}(x, \lambda), \sup_{\lambda \in \mathcal{S}_\lambda} \hat{\theta}^{lin}(x, \lambda) \right)$

...where $\hat{\theta}^{lin}(x, \lambda) := \hat{\theta}^{lin}(x, \lambda_0) + \frac{\partial \hat{\theta}^{lin}(x, \lambda)}{\partial \lambda} \Big|_{\lambda_0} (\lambda - \lambda_0)$.

Many variants are possible!

- Cross-validation [Giordano et al., 2019b]
- Prior sensitivity in Bayesian nonparametrics [Giordano et al., 2021]
- Model sensitivity of MCMC output [Giordano et al., 2018]
- Frequentist variances of MCMC posteriors (in progress)